

Switches, MAC addresses and ARP

Objective

1. Be able to construct and test a simple switched network
2. Understand how MAC addresses are allocated and how to find the MAC addresses in a network
3. Have a better understanding of how ARP is used to learn MAC addresses
4. Understand how switches learn the MAC addresses of devices connected to them

Exercise 1. Introduction to Cisco Switches

Equipment required:

- 2x PC's
- 1x CISCO 2960 switch
- 2x Copper straight-through cables

1.1 Theory

A switch is a networking concentration device sometimes referred to as a multiport bridge. Switches are relatively inexpensive and easy to install. When operating in full-duplex mode, they provide dedicated bandwidth to workstations. Switches reduce collisions by creating microsegments between ports to which the two workstations are attached. They are appropriate for small to large LANs with moderate to heavy traffic.

In addition to the layer 1 physical and layer 2 data link connections the computers in a switched LAN should be configured with the correct IP network settings, which is Layer 3 (network), so that they can communicate.

Take a look at the Switches in the device palette on *Cisco Packet Tracer*, find the Cisco 2960 switch. Drag an instance of this switch on to the work area. Once you've done this, click on the device and look around at the various properties of the device. See if you can identify the connections etc., on the front and back of the switch. You may want to see what you can learn from the web about them. If you have any problems, please check with your practical demonstrator.

Once you've done this, complete the network by dragging two PC's from the device palette on to the work area.

Documentation Task 1.

State what features you can see on the switches (ports, power switches, etc) and what you know of their functions.

Manually set the IP addresses

We are now going to tell the computers what the addresses they are going to have for the little network you are going to create. In most real networks you would select the option *Obtain an IP address automatically*. However, in this exercise we simply connecting the machines to each other, so they have nowhere they can obtain an address, so we have to set it manually.

In *Packet Tracer*, go to the PC's *Config* tab and enter the following configuration there (make sure you rename the first PC you create in Packet Tracer so that it isn't called *PC0* — if you don't the addressing scheme that follows will cause you problems).

The configuration provides the machine its IP address, what sort of subnet it is in and how it can communicate with the internet (note the last is not relevant for this exercise). Often, we will use some logical scheme to allocate addresses to machines. Give each of your PCs the *IP address 192.168.1.x* where *x* is the number of your PC. Eg. For *PC2* the address would be *192.168.1.2*. Note that addresses must be unique, if we gave them the same address it would cause a lot of confusion. Make the *Subnet mask 255.255.255.0*. We could leave the subnet mask alone, but the addresses we've given the machine are class C IP addresses, so let's do this properly. Leave all the other fields alone (they are probably blank).

1.2 Connecting the Network

1. Connect the computers to the switch using the copper straight-through cables. A copper straight-through cable must run from the NIC outlet on the blue panel corresponding to your PC to one of the outlets on the left of the switch.
2. Look at the link lights on the switch above the ports you plugged the cables into. Notice that the link lights on the switch go green (at least that's what they will do if everything is working correctly).

Documentation Task 2.

What IP address has been assigned to each computer?

What switch port is each computer connected to?

Exercise 2. Checking the Network

You should now check that your computers can communicate over your network. One way to do that is using ping. We will cover ping in more detail using Wireshark in later weeks.

Documentation Task 3.

Document what system tool you used to check whether your network was working and what the results were.

Exercise 3. Introduction to MAC Addresses

In this exercise we are going to learn a little more about switches and how they get messages to the proper destination.

IP addresses are layer 3, according to the Internet (and OSI model for that matter). A switch is actually a layer 2 (data link) device. It does not know how to send packets using IP. However, devices in LANs have an IP address. You used them when you used *ping* to check the connection between computers last week. So, if the switch does not know about IP, how is the message delivered to the correct recipient?

The first part of the puzzle are the MAC (Medium Access Control) addresses.



At this stage you can consider what is the difference between the layer 2 data-link layer and the layer 3 network layer. A layer 3 address is like buying a train ticket at Macquarie University to go to Central. The ticket just gives the end points of the journey. Likewise in the network layer, the IP addresses just deal with the original source host and final destination host of the message.

By contrast the data-link layer is concerned with all the hops, like an itinerary. It sends the train via Macquarie Park, Chatswood, Artarmon, St Leonards, etc until it reaches Central. The MAC addresses are only relevant to a single hop, so at Macquarie University, the MAC address of the next called at station is Macquarie Park, then from Macquarie Park to Chatswood, etc. On each hop, the data-link layer header changes to reflect the MAC address of the next hop (station); however, the network-layer header never changes – the IP address of the final destination of Central is always the same (the destination on your ticket). You could also choose to travel to Central along different routes via Epping and Strathfield, or Epping and Hornsby on the same ticket. This is the topic of routing, which is also handled at the network layer, since it is a high-level decision made by the customer, not a low-level decision made by the railway.

Most topics in computer networking have similar real-world analogies and you can think of packages sent via the Post Office.

MAC addresses are 48 bits long, usually specified in hexadecimal notation. For example:

0561-8c01-05b2 or 0561.8c01.05b2 or 05:61:8c:01:05:b2

In this format each hex digit represents **4 bits**. With some devices, the **12 hex characters** may be displayed as three sets of four characters separated by **periods or colons** (0002.A59A.635C). Hexadecimal is base 16 and uses the digits 0,1,2,3,4,5,6,7,8,9,a,b,c,d,e,f. Look it up if you haven't come across it before.

To find the MAC address of your PC, open up a command prompt window and type `ipconfig /all`

In Packet Tracer:

- Open PC → Desktop → Command Prompt
- Run: `ipconfig /all`

Please note that there is space between 'config' and '/'.

Record:

- IP Address
- MAC Address (Physical Address)

Documentation Task 4.

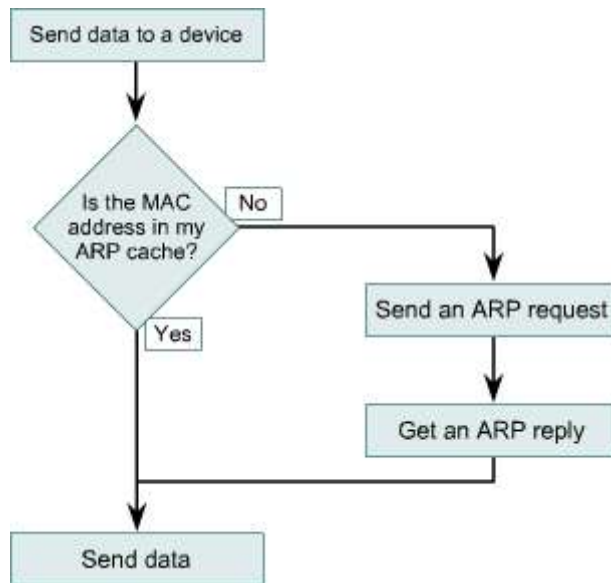
Include in your documentation what you consider to be the useful information from `ipconfig /all` for Ethernet connection on one of the PC's in Packet Tracer.

For the PCs we are using in the practical sessions you may see one a few addresses of the type above. Each network connection device has its own MAC address – and these PCs have multiple connection devices. Hence, multiple MAC addresses. It does not matter what you do with the computers, those MAC address are constant. This is independent of how you connect the machine to a network, if you change the switch the computer is connected to, etc. Even if the allocated IP address changes, the MAC address remains the same.

Exercise 4: Introduction to ARP (Address Resolution Protocol)

ARP is used to find what Media Access Control (MAC) Layer 2 address corresponds to a given IP (network Layer 3) address. The TCP/IP network protocol relies on IP addresses (eg., `192.168.14.211`) to identify individual devices and to assist in navigating data packets between networks. While the IP address is essential to move data from one LAN to another, it cannot deliver the data in the destination LAN by itself. The connections in LANs work at layer 2 (switches are layer 2 devices). Local network protocols, like Ethernet or Token Ring, use the MAC, or Layer 2, addresses to identify local devices and deliver all data.

A table is maintained in the computer of IP and MAC address combinations (a DHCP enabled switch maintains this information as well). In other words, it keeps track of which MAC address is associated with an IP address. If ARP does not know the MAC address of a local device, it issues a broadcast using the IP address. This broadcast searches for the MAC address that corresponds to the IP address. If the IP address is active on the LAN, it will send a reply from which ARP will extract the MAC address. ARP will then add the address combination to the local ARP table of the requesting computer.



MAC addresses and therefore ARP are only used within the LAN. When a computer prepares a packet for transmission, it checks the destination IP address to see if it is part of the local network. It does this by checking to see if the network portion of the IP address is the same as the local network. If it is, the ARP process is consulted to get the MAC address of the destination device using the IP address. The MAC address is then applied to the data packet and used for delivery.

If the destination IP address is not local, the computer will need the MAC address of the default gateway. The default gateway is the router interface that the local network is connected to in order to provide connectivity with other networks. The gateway MAC address is used because the packet will be delivered there and the router will then forward it to the network it is intended for.

If the computer does not receive any packets from an IP address after a few minutes, it will drop the MAC/IP entry from the ARP table assuming the device has logged off. Later attempts to access that IP address will cause ARP to do another broadcast and update the table.

Documentation Task 5.

Let's have a look at the ARP table held by the PCs. Get a command window running (if you do not already have one). In the window type **arp -a** and press Enter.

What devices have an entry in the ARP table? Record this in your documentation.

Could you see the physical address of the other PC(s) and the switch in your network? If not, ping the other computer(s) and try **arp -a** again. Try all the IP addresses you found in use in your network. Are entries for them now in the table? Add the results to your documentation if necessary. Do any IP addresses share the same MAC address?

A couple of other things to try:

arp -? to see the options and help for the arp command

ping -? to see the options and help for the ping command

Exercise 5. Examining ARP

Remember that the ARP protocol typically maintains a cache of IP-to-Ethernet address translation pairs on your computer in the switch. The *arp* command (in both MSDOS and Linux/Unix) is used to view and manipulate the contents of this cache. Since the *arp* command and the ARP protocol have the same name, it's understandably easy to confuse them. But keep in mind that they are different - the *arp* command is used to view and manipulate the ARP cache contents, while the Address Resolution Protocol (ARP) defines the format and meaning of the messages sent and received, and defines the actions taken on message transmission and receipt. Let's take a look at the contents of the ARP cache on your computer:

- **MS-DOS.** The *arp* command is in `c:\windows\system32`, and the full table is displayed by *arp -a* so type either that or `c:\windows\system32\arp -a` in the MS-DOS command line.
- **Linux/Unix.** The executable for the *arp* command can be in various places. Popular locations are `/sbin/arp` (for linux) and `/usr/etc/arp` (for some Unix variants).

In order to observe your computer sending and receiving ARP messages, we'll need to clear the ARP cache, since otherwise your computer is likely to find a needed IP-Ethernet address translation pair in its cache and consequently not need to send out an ARP message.

- a. **MS-DOS/Windows.** The MS-DOS *arp -d ** command will clear your ARP cache. The *-d* flag indicates a deletion operation, and the *** is the wildcard that says to delete all table entries. Under MS-Windows you will need to run this command from an "elevated" command prompt; to do this right-click on the command prompt icon in the taskbar and select "Run as Administrator"
- b. **Linux/Unix.** The *arp -d ** will clear your ARP cache. In order to run this command you'll need root privileges. If you don't have root privileges and can't run Wireshark on a Windows machine, you can skip the trace collection part of this lab and just use the trace discussed in footnote 1.

In Packet Tracer:

1. Switch to Simulation Mode
2. Generate traffic using ping between PCs
3. Observe ARP packets in the Event List
4. Click on packets to inspect:
 - Source MAC
 - Destination MAC
 - ARP Request and Reply